

Linear Programming Report

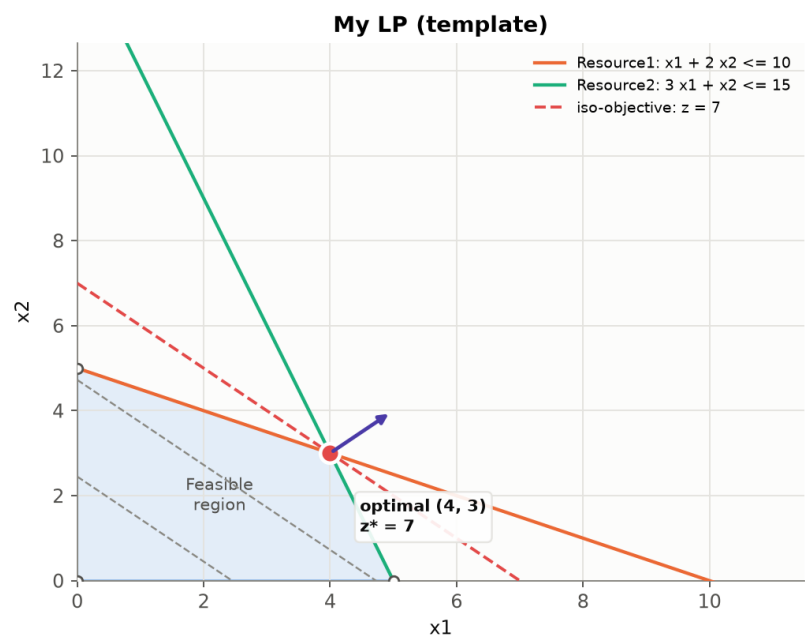
Sensitivity Analysis & Duality

My LP (template)

Template. sense: 'max' or 'min'. constraint sense: '<=' , '>=' or '='. variable sign: '>=0' , '<=0' or 'free'.

Source: Copy this file, edit it, and load it in the notebook

Problem type	MAX, 2 variables, 2 constraints
Status	OPTIMAL
Optimal Objective	7
Dual status	OPTIMAL
Prepared by	Dr. Islam Ali
Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	28 September 2026, 08:45



1. Model formulation

MAX $z = x_1 + x_2$
subject to
 $x_1 + 2x_2 \leq 10$ (Resource1)
 $3x_1 + x_2 \leq 15$ (Resource2)
 $x_1 \geq 0, x_2 \geq 0$

Data table

Constraint	x1	x2	Sense	RHS
Resource1	1	2	$\leq$	10
Resource2	3	1	$\leq$	15
Objective c	1	1	max	—

Definitions

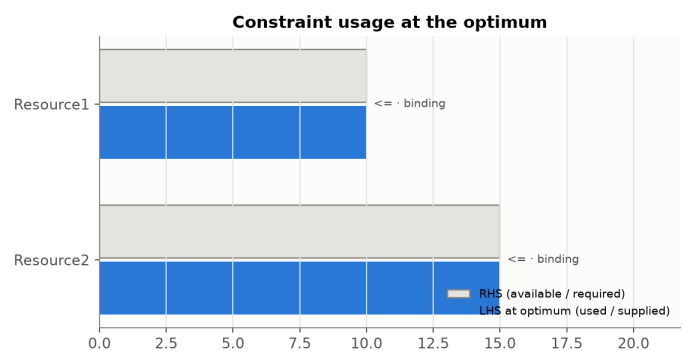
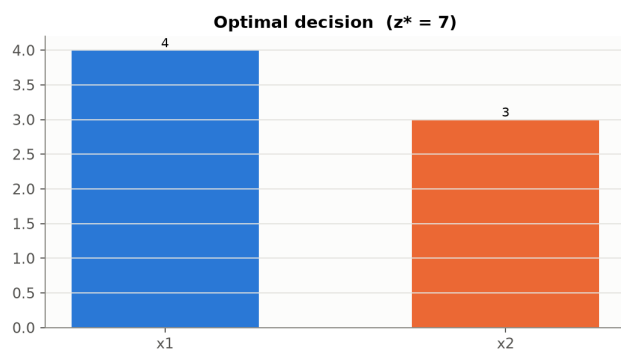
- **x1**: describe x1
- **x2**: describe x2
- **Resource1**: describe it
- **Resource2**: describe it

2. Optimal solution

Optimal Objective: 7

Variable	Description	Optimal value	Obj. coefficient	Contribution
x1	describe x1	4	1	4
x2	describe x2	3	1	3

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Resource1	10	<=	10	0	yes
Resource2	15	<=	15	0	yes



Corner points

x1	x2	Objective	Optimal?
0	0	0	
5	0	5	
4	3	7	✓
0	5	5	

3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
x1	4	0	1	2	0.5	0.5	3
x2	3	0	1	1	0.6667	0.3333	2

Right-hand-side ranging (constraints)

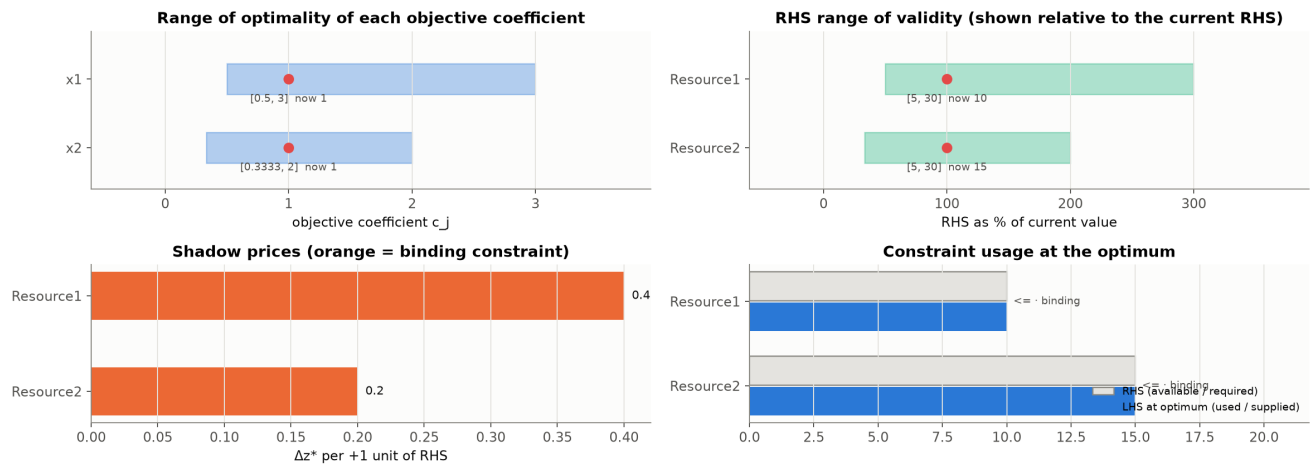
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Resource1	10	<=	10	0	0.4	20	5	5	30
Resource2	15	<=	15	0	0.2	15	10	5	30

Interpretation

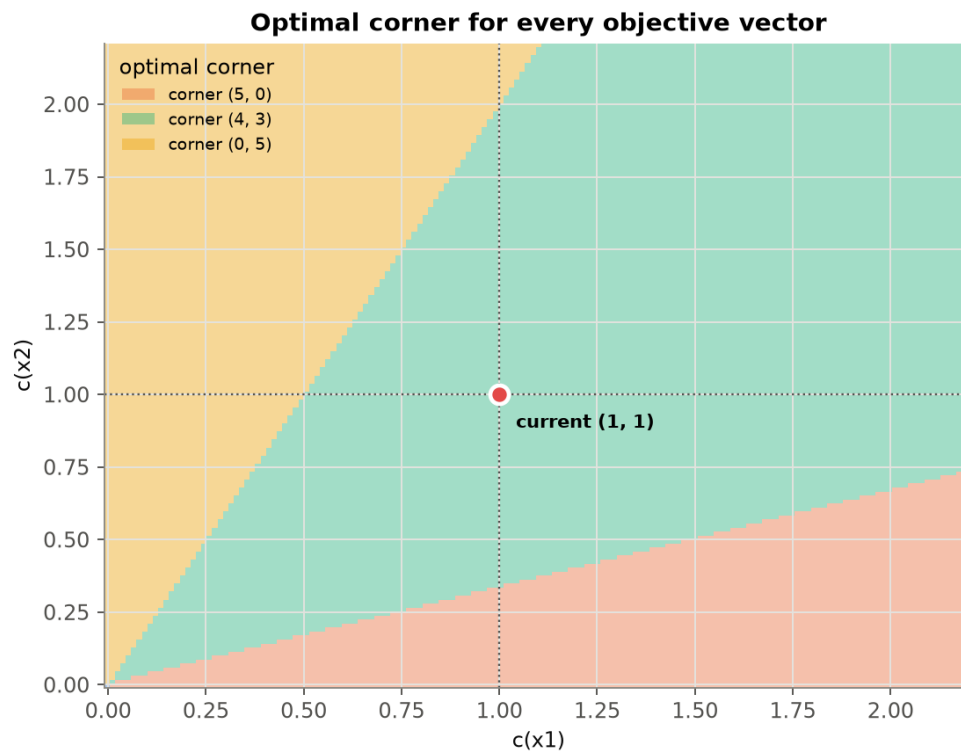
- **x1 = 4.** Its objective coefficient (1) may vary in [0.5, 3] without changing the optimal solution (one change at a time).
- **x2 = 3.** Its objective coefficient (1) may vary in [0.3333, 2] without changing the optimal solution (one change at a time).
- **Resource1** is binding. Shadow price 0.4: each extra unit of RHS changes Objective by 0.4, valid while the RHS stays in [5, 30].
- **Resource2** is binding. Shadow price 0.2: each extra unit of RHS changes Objective by 0.2, valid while the RHS stays in [5, 30].

Sensitivity dashboard

My LP (template): sensitivity dashboard ($z^* = 7$)

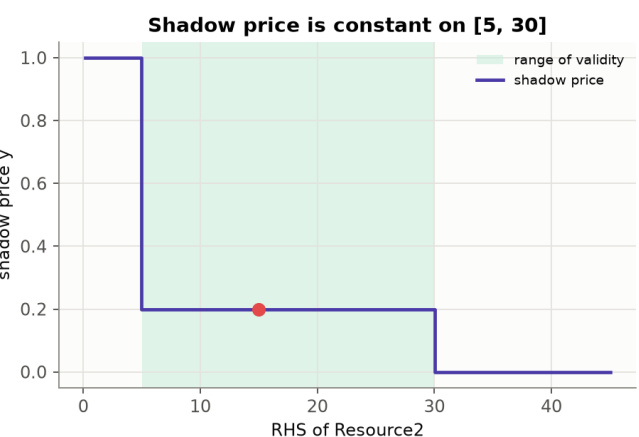
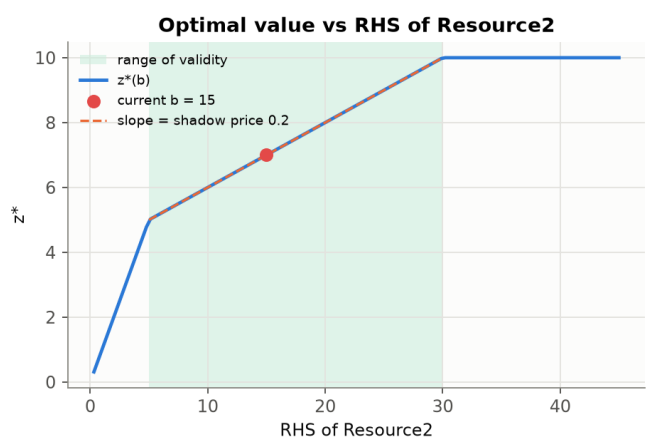
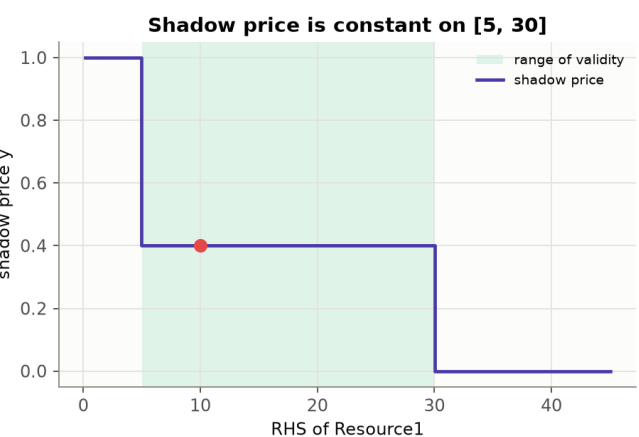
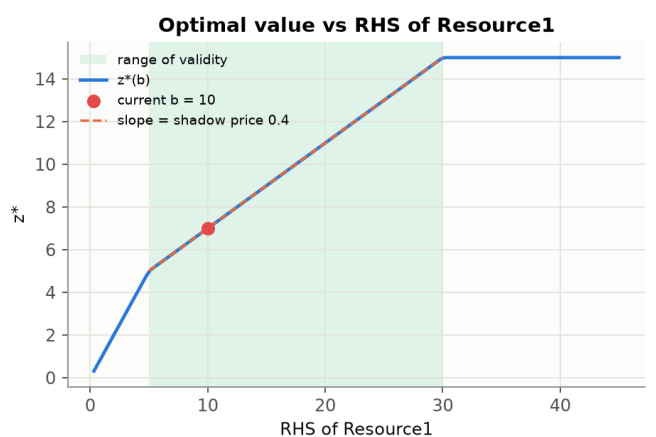
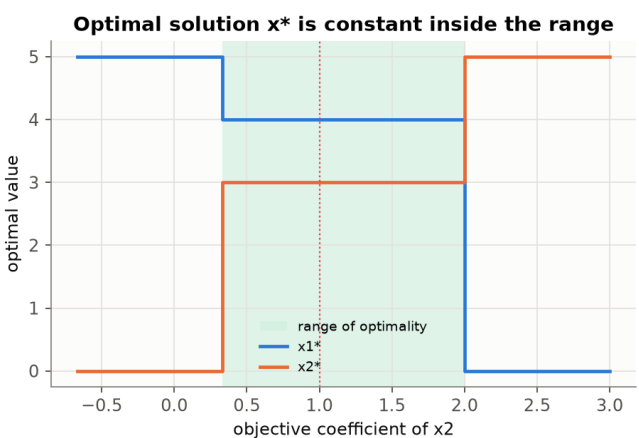
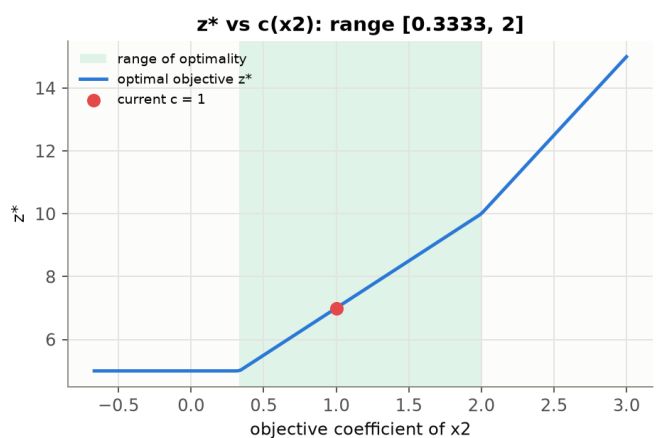
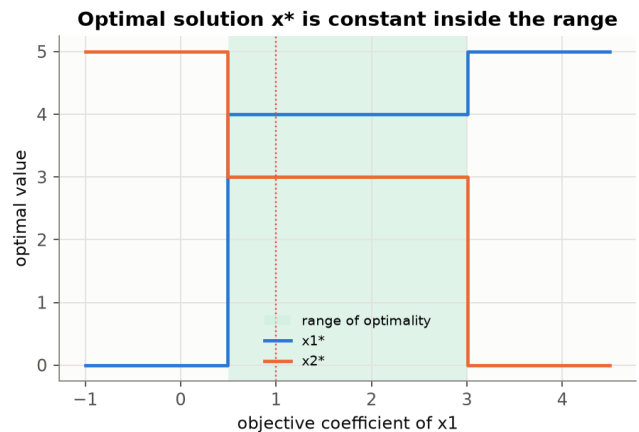
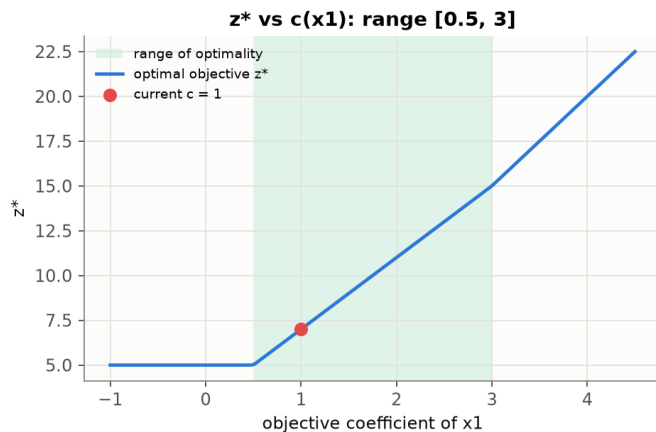


Optimal corner for every objective vector



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.

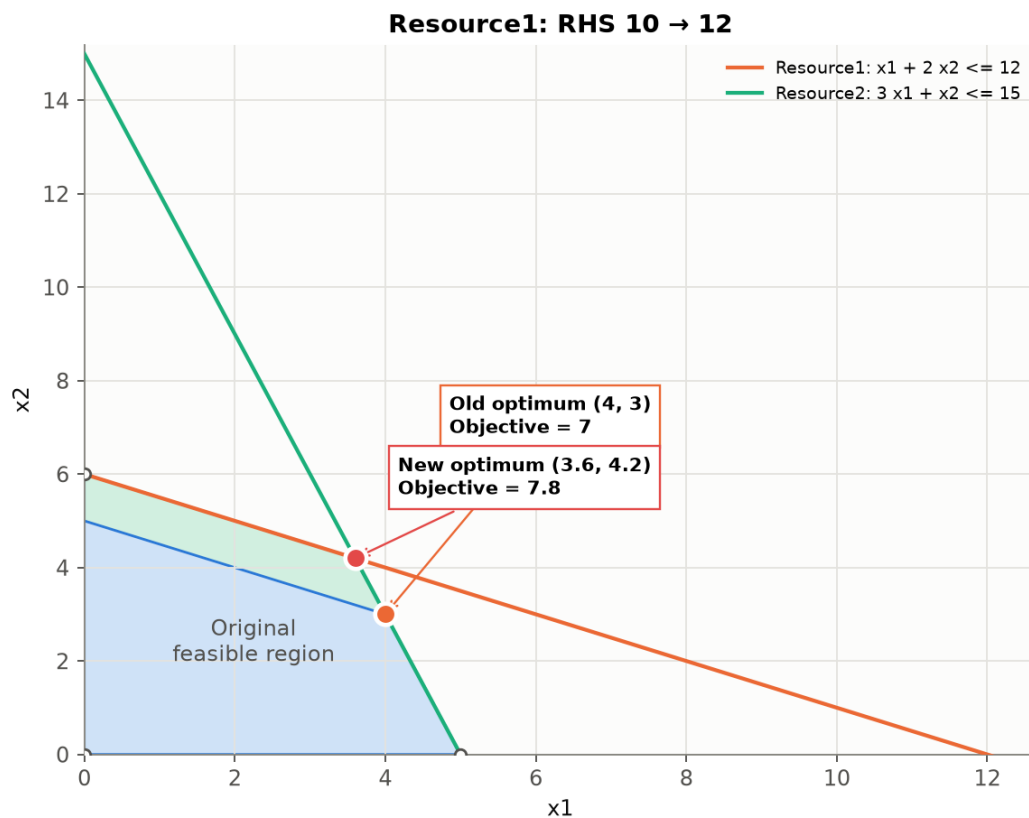
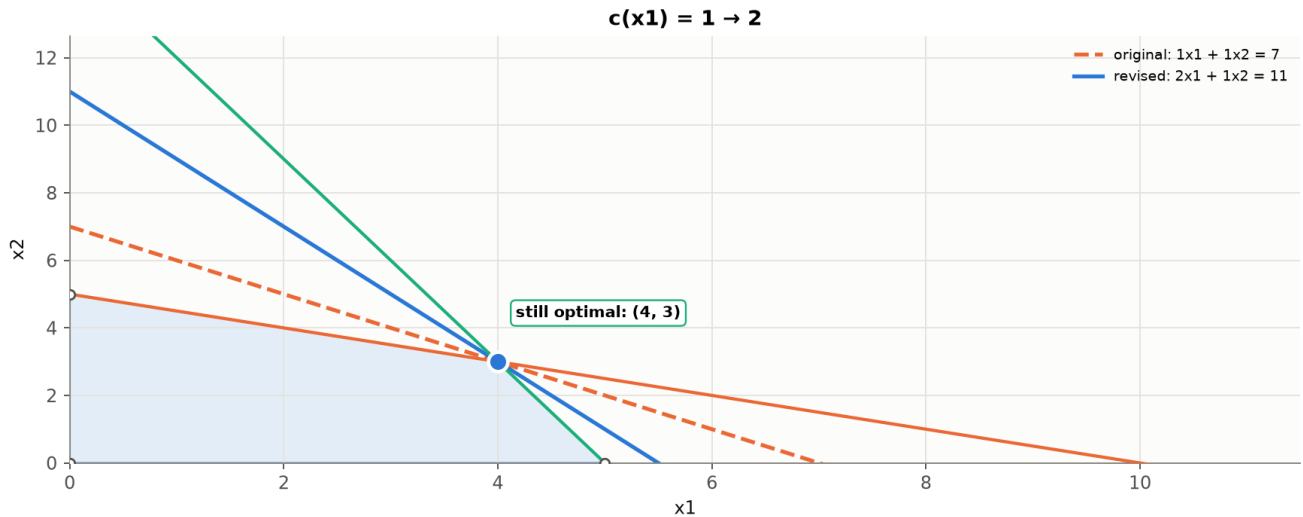


5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(x1): 1 → 2	yes	11	11	4, 3	what-if
b(Resource1): 10 → 12	yes	7.8	7.8	3.6, 4.2	what-if

Changing an objective coefficient rotates the iso-profit line; the feasible region never moves



6. Duality

Dual formulation

MIN $z = 10 \cdot y_{\text{Resource1}} + 15 \cdot y_{\text{Resource2}}$
 subject to
 $y_{\text{Resource1}} + 3 \cdot y_{\text{Resource2}} \geq 1 \quad (x1)$
 $2 \cdot y_{\text{Resource1}} + y_{\text{Resource2}} \geq 1 \quad (x2)$
 $y_{\text{Resource1}} \geq 0, y_{\text{Resource2}} \geq 0$

Dual solution (resource / requirement prices)

Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
$y_{\text{Resource1}}$	Resource1	0.4	0.4	4
$y_{\text{Resource2}}$	Resource2	0.2	0.2	3

Strong duality check: primal $z^* = 7$, dual $w^* = 7$ (equal ✓). Dual of the dual equals the primal: yes.

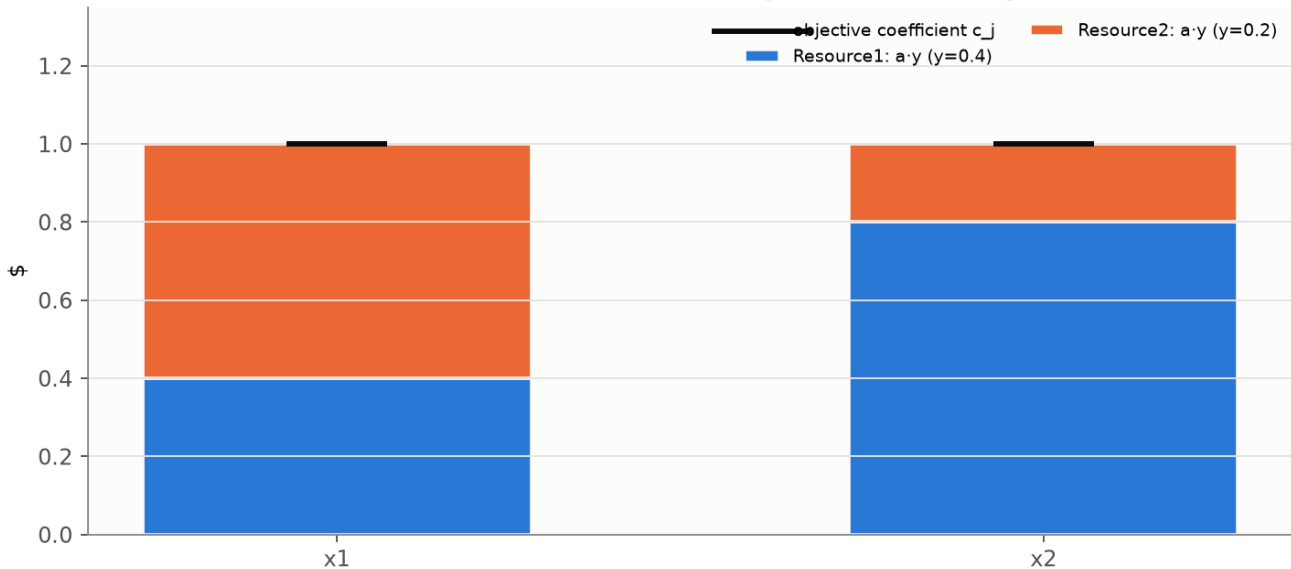
Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Resource1 slack $\times y_{\text{Resource1}}$	0	0.4	0
Resource2 slack $\times y_{\text{Resource2}}$	0	0.2	0
$x1 \times$ surplus of dual constraint $x1$	0	4	0
$x2 \times$ surplus of dual constraint $x2$	0	3	0

1 vs 1
tight

1 vs 1
tight

Dual constraints: value of resources in each product vs its objective coefficient

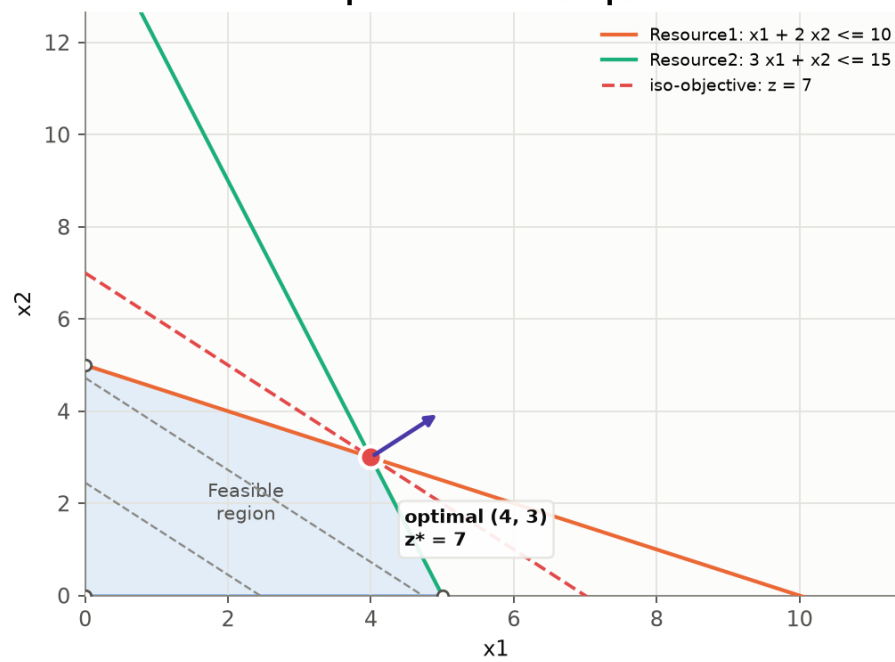


Primal (max): 2 constraints × 2 variables

1	1	← objective
1	2	≤ 10
3	1	≤ 15
x_1	x_2	

Dual (min): 2 constraints × 2 variables

10	15	← objective
1	3	≥ 1
2	1	≥ 1
$y_{\text{Resource1}}$	$y_{\text{Resource2}}$	

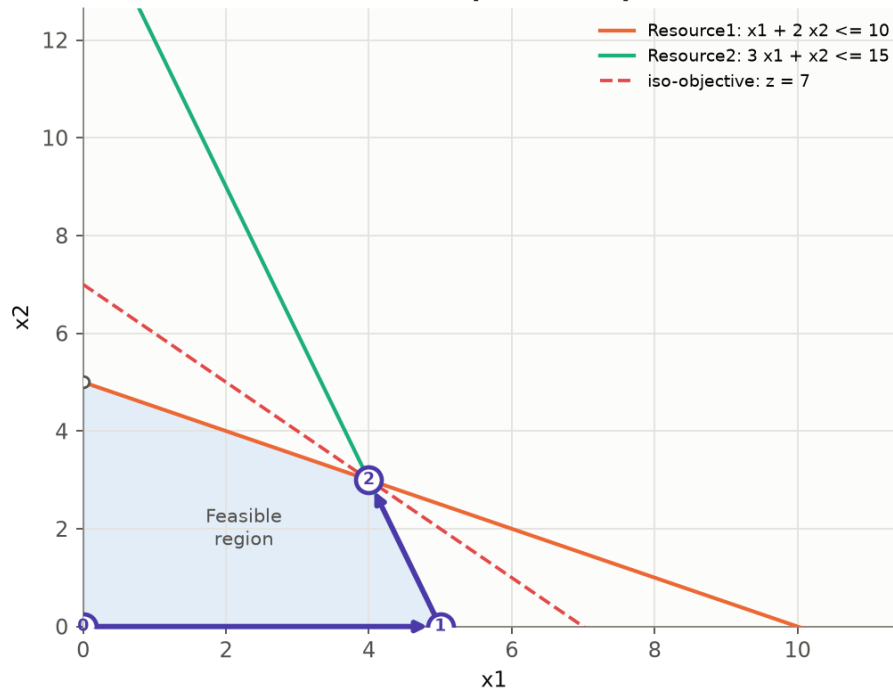
Graphical view of the primal

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_i - c_i$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	2	x1	s2	0	yes	no	(0, 0)
1	primal	2	x2	s1	5	yes	no	(5, 0)
2	primal	2			7	yes	yes	(4, 3)

Path of the primal simplex



Primal simplex - phase 2, tableau 0
enter x1, leave s2 (min ratio 5)

	x1	x2	s1	s2	RHS
z	-1	-1	0	0	0
s1	1	2	1	0	10
s2	3	1	0	1	15

Primal simplex - phase 2, tableau 1
enter x2, leave s1 (min ratio 3)

	x1	x2	s1	s2	RHS
z	0	-0.667	0	0.333	5
s1	0	1.67	1	-0.333	5
x1	1	0.333	0	0.333	5

Primal simplex - phase 2, tableau 2
optimal tableau

	x1	x2	s1	s2	RHS
z	0	0	0.4	0.2	7
x2	0	1	0.6	-0.2	3
x1	1	0	-0.2	0.4	4